



$$E = \hbar \omega$$

$$|E_1 - E_2| = \hbar \omega$$

$$\omega^2 = \frac{1}{m} (k_1 + k_2 + (k_1 + k_2))$$

$$\omega = \sqrt{\frac{k_1}{m}} \pm \sqrt{\frac{k_2}{m}}$$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} e^{-i(\omega_1 \pm \omega_2)t}$$

NORMAL MODES =
2 x UNIT CELLS
= 4 MODES



GENERAL

If M atoms per unit cell

M modes at each k

1. Acoustic

($M-1$) are optical

In 4-dimensionality

4M modes at each k

1. Acoustic

($M-1$) are optical

$$\omega^2 = \frac{1}{m} \left(\frac{1}{L} \int_0^L \sin^2 \frac{n\pi x}{L} dx \right)$$

$$1 \text{ or } 2 \text{ } \omega_n^2 = \frac{1}{m} \left(\frac{1}{L} \int_0^L \sin^2 \frac{n\pi x}{L} dx \right)$$

$$\omega^2 = \frac{1}{m} \left(\frac{1}{L} \int_0^L \sin^2 \frac{n\pi x}{L} dx \right)$$

$$\omega = \sqrt{\frac{1}{m} \left(\frac{1}{L} \int_0^L \sin^2 \frac{n\pi x}{L} dx \right)}$$

$$\left(\frac{1}{L} \int_0^L \sin^2 \frac{n\pi x}{L} dx \right) = \frac{1}{2} \left(1 + \frac{1}{n^2} \right)$$

NORMAL MODES =

2 x NORMAL MODES
= 2 MODES



GENERAL

1. M atoms of unit cell

M modes at each k

1 is acoustic

others are optical

in 4 dimensions

4M modes at each k

1 acoustic mode

others are optical

grew 1 inch 2 months
2 months 2 months 2 months
2 months 2 months 2 months

$\text{Wavelength} = \frac{2 \times \text{width of slit}}{\text{angle}}$

Ch. 11.11

1. *Chlorophyll a* (Chl a) is the primary photosynthetic pigment in most plants and algae. It is a green pigment that absorbs light energy in the blue and red regions of the visible spectrum. Chl a is essential for the light-dependent reactions of photosynthesis, where it converts light energy into chemical energy in the form of ATP and NADPH.

$$|K_1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$|K_2\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|K_3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$|K_4\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

EACH OF THE 2 QUANTUM MODES
 ω_0
 2 QUANTUM MODES =
 $2 \times \text{HUBBARD}$
 $\times \text{HUBBARD}$

$$\begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \times \frac{1}{\sqrt{2}}$$



GENERAL
 # OF ATOMS IN
 # MODES AT EACH
 1 = HUBBARD
 (HUBBARD) HAS HUBBARD
 IN 1-D HUBBARD
 1-D HUBBARD AT EACH

2-D HUBBARD
 2-D HUBBARD AT EACH





problem 1

$$H_0: \mu = \mu_0 \quad H_1: \mu > \mu_0$$
$$\alpha = \frac{1}{n} \left(\frac{1}{\sigma^2} \left(\frac{1}{n} \sum_{i=1}^n x_i^2 - \mu_0^2 \right) \right)$$
$$\rightarrow \sqrt{\frac{1}{n}} \cdot \sqrt{\frac{1}{\sigma^2}}$$

$$\left(\frac{1}{n} \sum_{i=1}^n x_i^2 - \mu_0^2 \right) \rightarrow \frac{1}{n} \sum_{i=1}^n x_i^2 - \mu_0^2$$

problem 2: with 2 samples

Bayesian model = 2 x model with 1 sample



problem 3: with 3 samples

Bayesian model = 3 x model with 1 sample





$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 \right)$$

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$$\frac{d}{dt} \left(\frac{1}{2} m v^2 \right) = \frac{d}{dt} \left(\frac{1}{2} m \dot{x}^2 \right)$$





$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -y \\ x \end{pmatrix}$$

$\Rightarrow$ all unit circles

... ..

$$x'' + y'' = 0$$
$$x' = \sqrt{1 - y'^2}$$
$$x = \int \sqrt{1 - y'^2} dy$$

... ..

$$x' = \sqrt{1 - y'^2}$$
$$x = \int \sqrt{1 - y'^2} dy$$

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$$x = \int \sqrt{1 - y'^2} dy$$





$$\frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{d^2}{dt^2} \right) = \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{d^2}{dt^2} \right) = \frac{1}{2} \frac{d}{dt} \left(\frac{1}{2} \frac{d^2}{dt^2} \right)$$

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$$f(x) = \frac{1}{2} \ln \left(\frac{x+1}{x-1} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{1}{x+1} - \frac{1}{x-1} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{x-1 - (x+1)}{(x+1)(x-1)} \right)$$

$$f'(x) = \frac{1}{2} \left(\frac{-2}{x^2-1} \right)$$

$$f'(x) = -\frac{1}{x^2-1}$$

$$R = \left(\frac{2\pi}{\omega} \right) \frac{1}{\sqrt{1 - \frac{\omega^2}{\omega_0^2}}}$$

$$\omega = 2\pi f$$

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$$f'(x) = -\frac{1}{x^2-1}$$



1940

$$K = \frac{[H_2O]}{[H_2][O_2]}$$

2000

[illegible]

19

一、

Figure 1

1. *Chrysomelids*





$$k = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

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$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$k = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\Rightarrow \text{symmetrisch}$$

$$\text{symmetrisch}$$

$$\text{symmetrisch}$$





$$y = -0.1x + 1.1$$

$$y = -0.1x + 1.1$$

$$y = -0.1x + 1.1$$

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$$y = -0.1x + 1.1$$

$$y = -0.1x + 1.1$$

$$y = -0.1x + 1.1$$



$$u = 10^{-10}$$

$$v = 10^{-10} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$u = 10^{-10}$$

$$v = 10^{-10}$$

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$$u = 10^{-10}$$

$$v = 10^{-10}$$

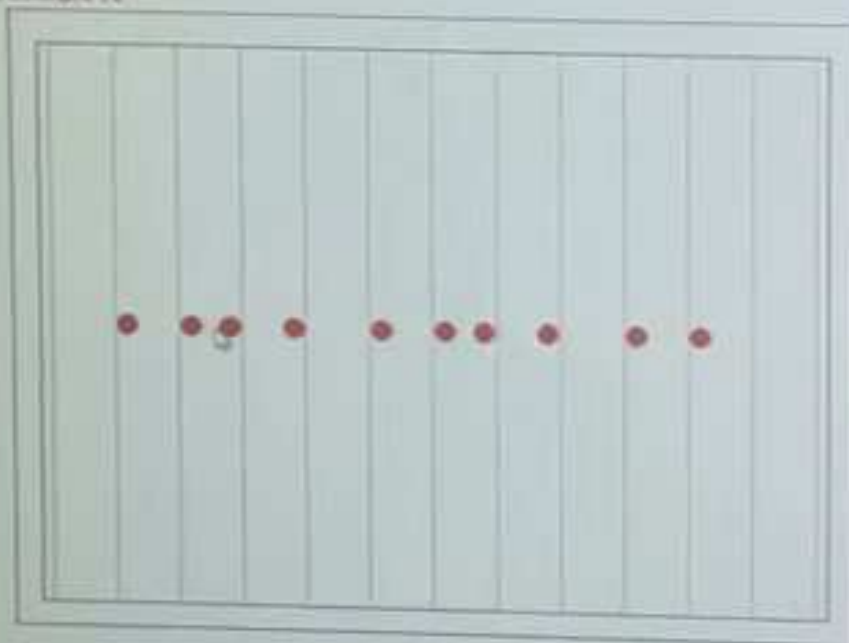
$$u = 10^{-10}$$

$$v = 10^{-10}$$

File Home Insert Drawing Tools Windows Help View

The length is the same for all the following points

The width is 2 x



Grid Drawing & Value

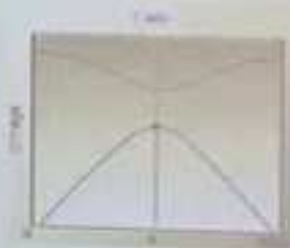


1	2	3
4	5	6
7	8	9
10	11	12

13

14

15



16

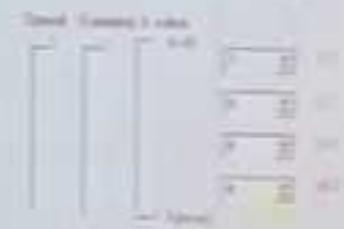
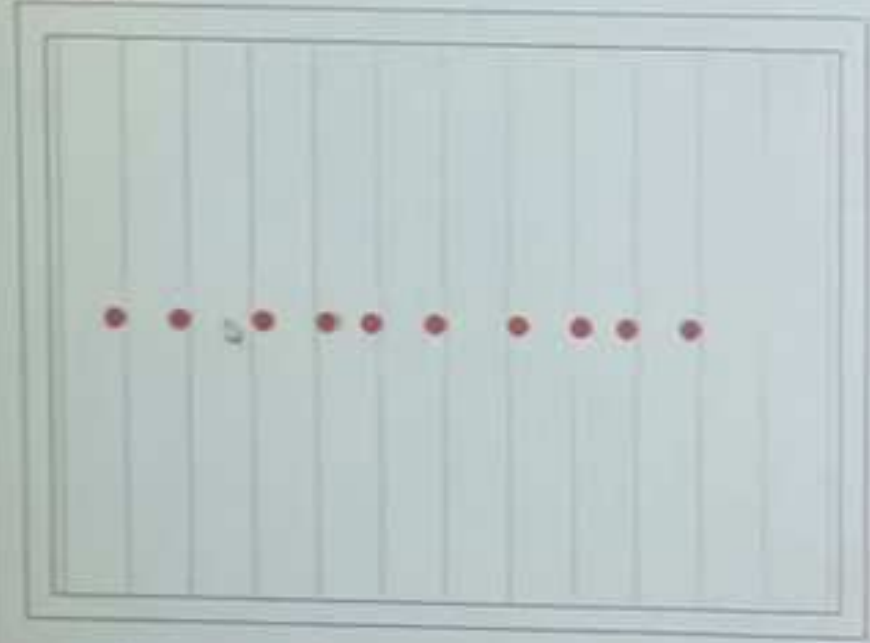
17

18

19

20

The length is the same the distance between them
 The wavelength is 2λ

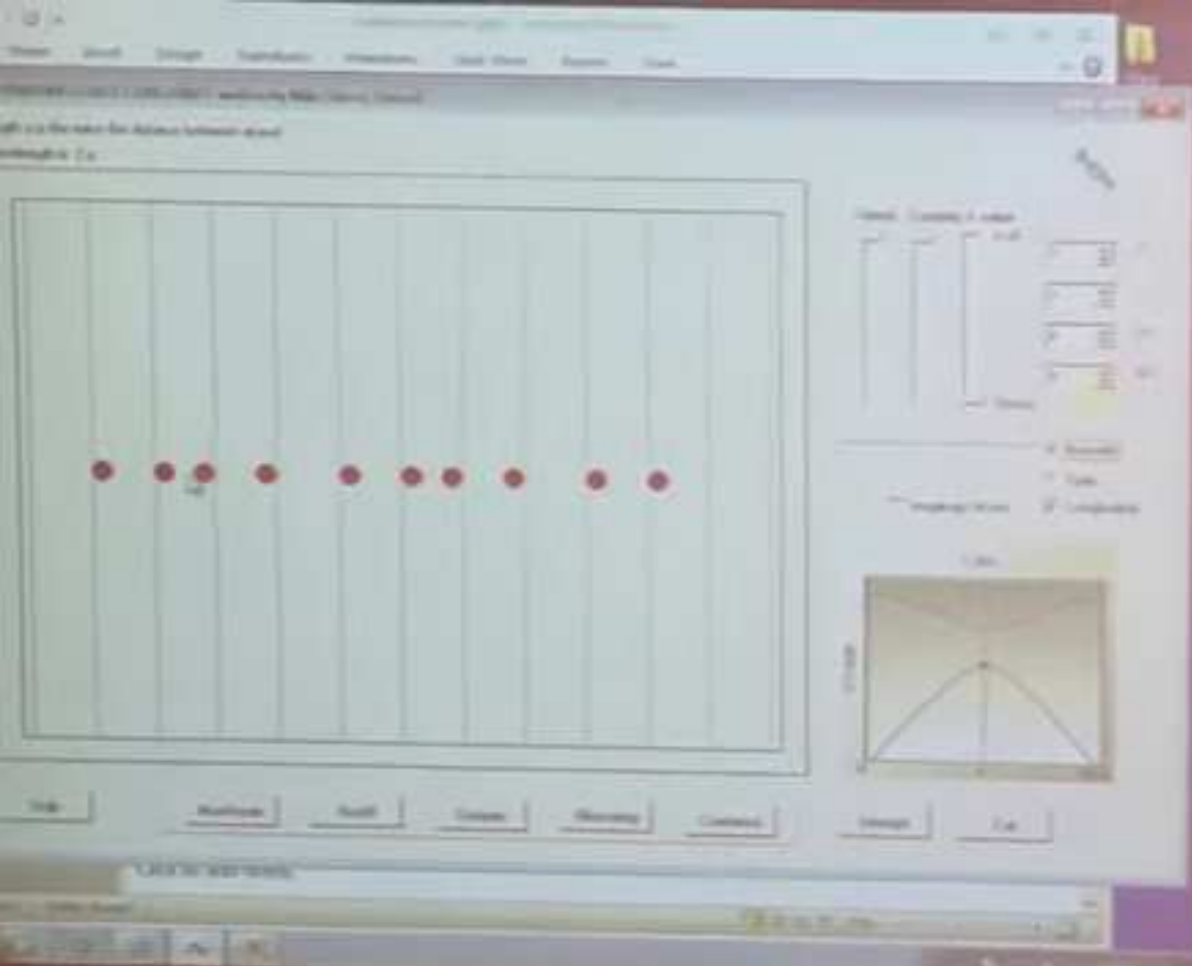


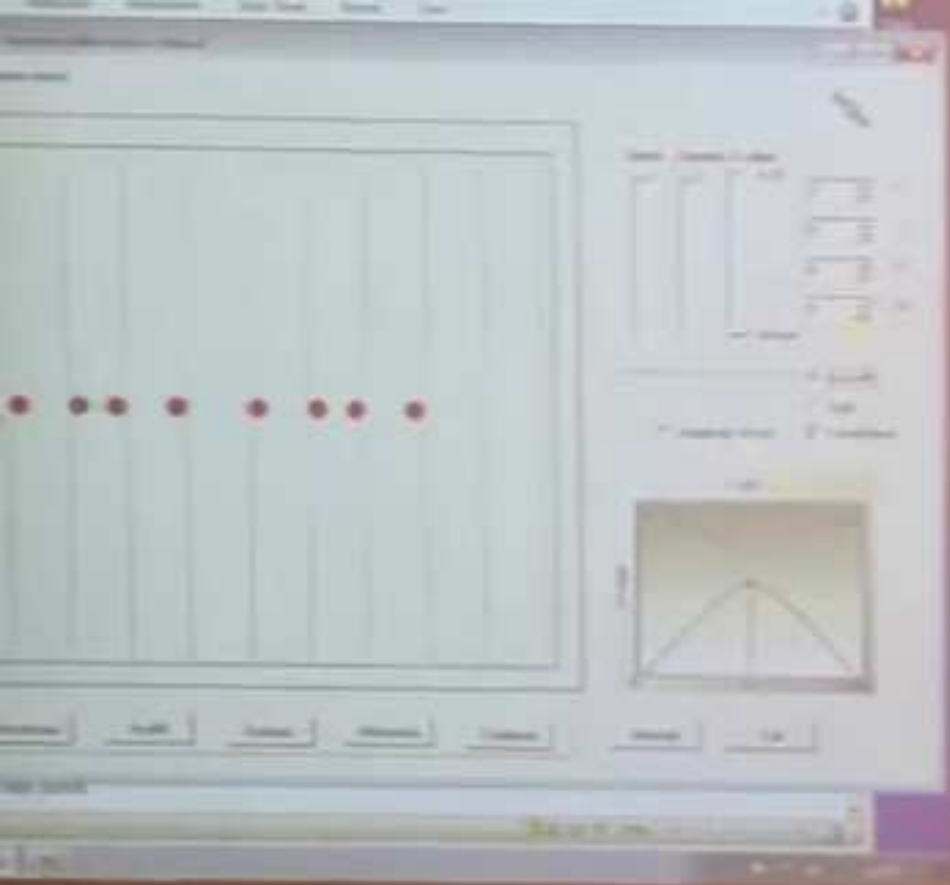
Graph type: ☐ Displacement vs. Time ☒ Displacement vs. Position



Graph type: ☐ Displacement vs. Time ☒ Displacement vs. Position

CREATING WAVES





$$m^2 = k_1 + k_2 = 15$$

$$\text{at } x=0$$

$$y(0) = -\frac{1}{2} \left(\frac{1}{2} \right)$$

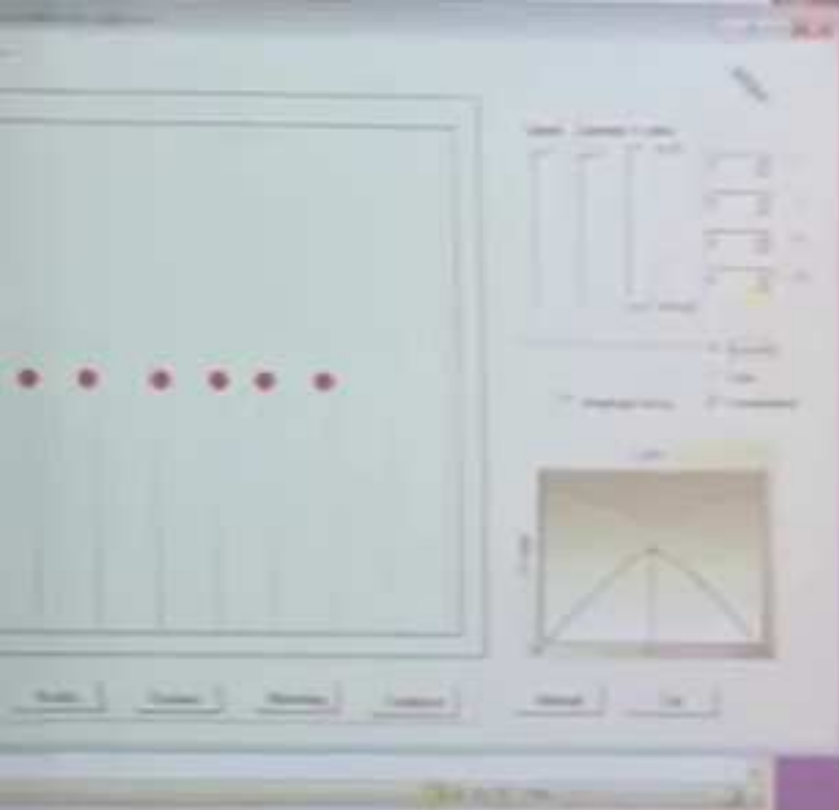
$$\frac{1}{2}$$

$$\left(\frac{1}{2} \right) - \left(\frac{1}{2} \right)$$

$$\frac{1}{2}$$

$$\left(\frac{1}{2} \right) - \left(\frac{1}{2} \right)$$

$$15 = k_1 + k_2 = 15$$



$$x_1^2 = x_1 + x_2 \quad | (x_1 - x_2) \cdot e^{-x_1}$$

$$\subseteq K = 0$$

$$p(x_1, x_2) = -f(x_1, x_2) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

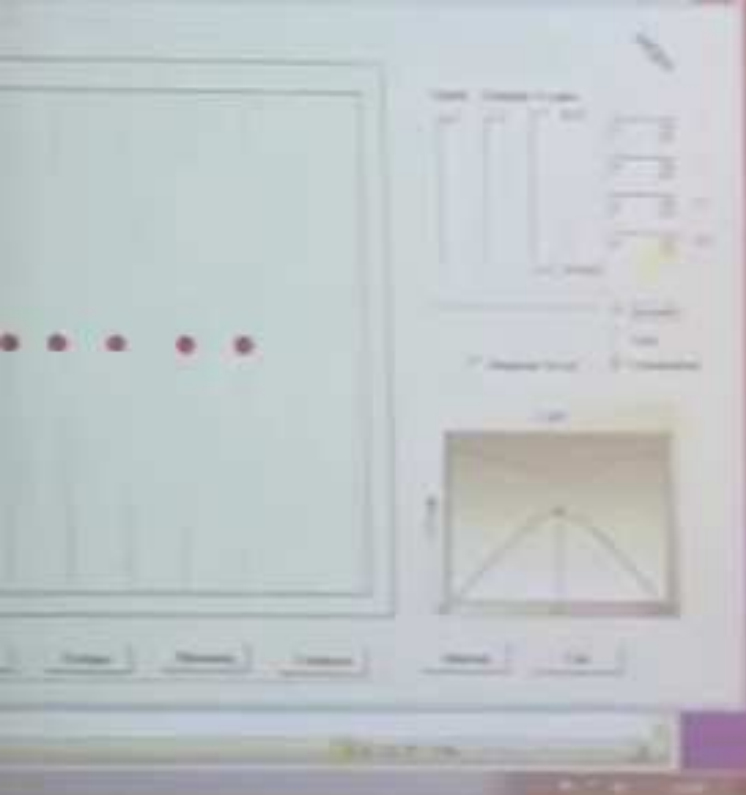
$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\text{Covarianz}$$

$$\text{Cov}(x_1, x_2) = \frac{1}{n} \sum_{i=1}^n (x_{1i} - \bar{x}_1)(x_{2i} - \bar{x}_2)$$

$$x_1 = \bar{x}_1$$

$$(x_1 - \bar{x}_1) \cdot (x_2 - \bar{x}_2)$$



$$x_{n+1} = x_n + \Delta x_n \approx |x_n - x_{n-1}| e^{-\Delta x_n}$$

$$\in \mathbb{R}^n$$

$$x_{n+1} = - (x_n - x_{n-1}) \begin{pmatrix} x_n - x_{n-1} \\ x_n - x_{n-1} \end{pmatrix}$$

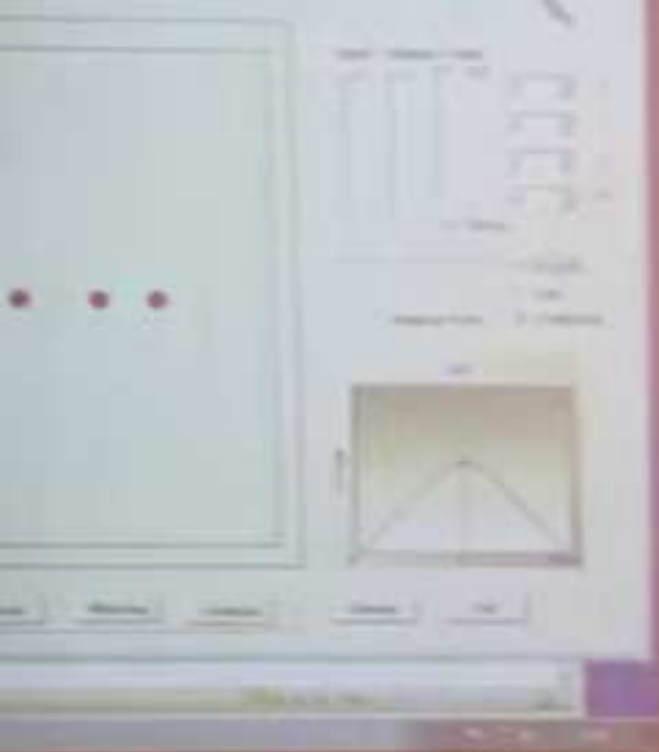
$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$x_{n+1} = x_n + \Delta x_n$$

$$x_{n+1} = x_n + \Delta x_n$$

$$x_{n+1} = \frac{1}{2} (x_n + x_{n-1})$$



$$E_k = 0$$

$$p(\mathbf{x}) = -(\mathbf{x} - \mathbf{x}_0) \begin{pmatrix} k_1 - k_2 \\ -k_1 + k_2 \end{pmatrix}$$

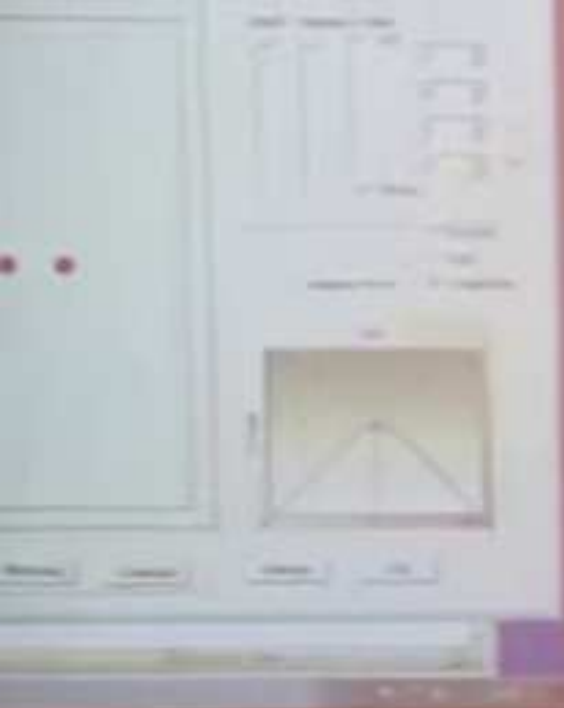
$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$E_k = 0$$

$$E_k = 0$$

$$E_k = \frac{1}{2} (k_1 + k_2 + (k_1 - k_2))$$



$$C = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$x(0) = - (x_1 - x_2) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$x(0) = - (x_1 - x_2) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$x(0) = - (x_1 - x_2) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$x(0) = - \frac{1}{2} (x_1 - x_2) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



$$C \approx 10$$

$$f(x) = -\frac{1}{2}(x-1)^2 + \frac{1}{2}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Consider
for simplicity
in 2D

$$f(x) = \frac{1}{2}x^2 + \frac{1}{2}x + \frac{1}{2}$$

$$f(x) = \frac{1}{2}(x^2 + x + 1)$$

$$= \sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}$$

$$k = \left(\frac{2\pi}{\lambda} \right)$$

DIFFERENT

or

$$= 0$$

Consider k

Consider also

$$2 \times \pi$$

$$L < 0$$

$$x_{\text{max}} = -\frac{1}{2} \left(\frac{1}{2} \right)$$

$$\frac{1}{2} \left(\frac{1}{2} \right)$$

$$\frac{1}{2} \left(\frac{1}{2} \right)$$

$$k = \left(\frac{2\pi}{L} \right) f_{\text{max}}$$

$$\bullet \text{ DIFFERENT } \frac{f_{\text{max}}}{k} = \frac{f_{\text{max}}}{k}$$

$$= \bullet \text{ OF } W$$

normal
mode
in the

$$x_1 = x_2 = x_3 = x_4$$

$$x_1 = x_2 = x_3 = x_4$$

$$\sqrt{\frac{1}{2}}, \sqrt{\frac{1}{2}}$$

EACH E HAS 2

normal modes =

$$2 = \bullet \text{ OF } W$$

$$= \bullet \text{ OF } W$$

$$\sum k = 0$$

$$f(x,y,z) = -f(x,y,z) \begin{pmatrix} 1 & -2 \\ -1 & 1 \end{pmatrix}$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lim_{x \rightarrow 0} \left(\frac{1}{x} \right) = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$k = \left(\frac{2\pi}{L} \right) p \quad p \in \mathbb{Z}$$

$$\text{Different } k_1 = \frac{(2\pi y/a)}{(2\pi/L)} = \frac{L}{a}$$

= # OF UNIT CELLS

PROBLEM
12. Find the
value of

$$(x_1 - x_2)^2 + (y_1 - y_2)^2$$

$$= \frac{1}{2} (x_1 + x_2 + i(y_1 + y_2))$$

$$= \sqrt{\frac{x_1^2}{2}} + \sqrt{\frac{y_1^2}{2}}$$

EACH K HAS 2 NORMAL MODES

2 NORMAL MODES =
2 + 2 UNIT CELLS
= 4 MODES

$$S \approx 0$$

$$H_{eff} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\frac{1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Consider
the system
with

$$(K_1 + K_2) \psi = (E_1 + E_2) \psi$$

$$\psi = \frac{1}{\sqrt{2}} (K_1 + K_2 + (K_1 + K_2))$$

$$\psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$k = \left(\frac{2\pi}{L} \right) \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ for } p \in \mathbb{Z}$$

$$\text{different } \frac{(2\pi/a)}{(2\pi/L)} = \frac{L}{a} = N$$

= # OF UNIT CELLS

EACH K HAS 2 adjacent
modes
by

adjacent modes =

2 adjacent cells
= 4 modes

$$E \approx 0$$

$$H_{\text{eff}} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$H_{\text{eff}} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$H_{\text{eff}} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$H_{\text{eff}} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$H_{\text{eff}} = -J(\sigma_x) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$k = \left(\frac{2\pi}{L} \right) p \quad p \in \mathbb{Z}$$

$$\# \text{ different } k = \frac{(2\pi/a)}{(2\pi/L)} = \frac{L}{a} = N$$

= # OF UNIT CELLS

EACH k HAS 2 NORMAL MODES

2 NORMAL MODES =

2 x UNIT CELLS
= 2 MODES

$$\Delta E = 2$$

$$H_{\text{matrix}} = - (H_1, H_2) \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\rightarrow \sqrt{\frac{H_1}{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\rightarrow \sqrt{\frac{H_2}{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

normal modes
 $H_1 = H_2$
 $\rightarrow H_1$

$$(H_1 + H_2) = (H_1 + H_2)$$

$$\omega^2 = \frac{1}{2M} (H_1 + H_2 + H_2 + J)$$

$$\omega = \sqrt{\frac{H_1}{2M}} + \sqrt{\frac{H_2}{2M}}$$

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$k = \left(\frac{2\pi}{L} \right) r_{\text{K}} \quad p \in \mathbb{Z}$$

$$\text{DIFFERENT } E = \frac{(R_{\text{A}}/a)}{(R_{\text{A}}/L)} = \frac{L}{a} = N$$

= # OF UNIT CELLS

EACH E HAS 2 NORMAL MODES

$E_{\text{NORMAL MODES}} =$

2 x normal cells
 = 4 modes

$$\omega = \sqrt{\frac{H_1}{2M}} + \sqrt{\frac{H_2}{2M}}$$

$$CL=0$$

$$H_{\text{min}} = -(\ln 1/2) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\rightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$k = \left(\frac{2\pi}{L} \right) p \quad p \in \mathbb{Z}$$

$$\text{DIFFERENT } k = \frac{(2\pi/L)}{(2\pi/L)} = \frac{L}{2} = N$$

= # OF UNIT CELLS

UNIFORM
OR VIBRATING
IN Y_0

$$(Y_1 - Y_0) = (Y_2 - Y_1)$$

$$Y = \frac{1}{2} (Y_1 + Y_2 + Y_3 + \dots)$$

$$\rightarrow \sqrt{\frac{Y_1}{2}} + \sqrt{\frac{Y_2}{2}}$$

EACH k HAS 2 NORMAL MODES
 ω_1

2 NORMAL MODES =

2 = # OF UNIT CELLS
2 = # OF MODES

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} \frac{1}{\sqrt{2}}$$



$$H(k) = -\hbar^2 \nabla^2 \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

CHARGE
PER SQUARE
IN V_0

$$E_1 = \frac{1}{2} E_2 = \frac{1}{2} E_3 = \frac{1}{2} E_4$$

$$\omega^2 = \frac{1}{m} (k_1 + k_2 + k_3 + k_4)$$

$$\omega = \sqrt{\frac{k_1}{m}}, \sqrt{\frac{k_2}{m}}$$

$$\left(\frac{1}{\mu} \right) = \left(\frac{1}{\mu_0} \right) = \frac{1}{\mu_0} \frac{1}{\mu_r}$$

$$k = \left(\frac{2\pi}{L} \right) p \quad p \in \mathbb{Z}$$

$$\# \text{ DIFFERENT } k = \frac{(2\pi/a)}{(2\pi/L)} = \frac{L}{a} = N$$

= # OF UNIT CELLS

EACH k HAS 2 NORMAL MODES
 ω_p

NORMAL MODES =
2 x UNIT CELLS
= 2N MODES



Acoustic mode

$\omega \sim |k|$ at

optical mode

$\omega \sim \sqrt{k^2 + k_0^2}$

at $k=0$

GENERAL

of atoms in

1D chain is N

1 = 2N modes

(N-1) acoustic

1 = 2N - 1 modes

1D chain is

$$P_{\text{matrix}} = -(\hat{p}_1 \hat{p}_2) \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$$

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\# \text{ DIFFERENT } k_1 = \frac{(2\pi/a)}{(2\pi/L)} = \frac{L}{a} = N$$

= # OF UNIT CELLS

ACOUSTIC MODE

$\omega \propto |k|$ AT LOW

OPTICAL MODE

$\omega \propto \sqrt{k^2}$ AT HIGH

ACOUSTIC
OR OPTICAL
MODES

$$(\omega_1 - \omega_2) = \omega_1 - \omega_2$$

$$\omega^2 = \frac{1}{2} (k_1^2 + k_2^2 + |k_1 - k_2|^2)$$

$$\omega = \sqrt{\frac{1}{2} (k_1^2 + k_2^2 + |k_1 - k_2|^2)}$$

EACH k HAS 2 NORMAL
MODES
 ω_1
 ω_2

NORMAL MODES =
2 x UNIT CELL
x # CELLS

ACOUSTIC

OF ACOUSTIC MODES

OF OPTICAL MODES

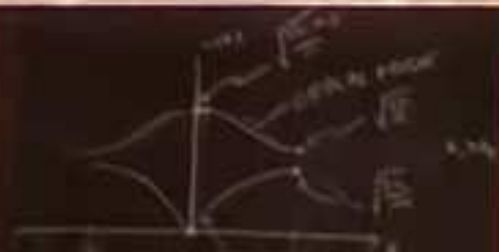
1 x # CELLS

(# OF) OPTICAL MODES

IN 2-DIMENSIONAL

THE MODES ARE 2 x

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} A_1 \\ A_2 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$k_1 = \frac{2\pi}{\lambda} = \frac{2\pi}{a} = N$$

= # OF UNIT CELLS

ACOUSTIC MODE

$\omega(k)$ AT $k=0$

OPTICAL MODE

UNBOUNDED
AC-COUSTIC

UNBOUNDED
AC-COUSTIC
AT $k=0$

$$|k_1 - k_2| = |k_1 + k_2|$$

$$\omega = \frac{1}{2} (k_1 + k_2 + |k_1 - k_2|)$$

$$\omega = \sqrt{\frac{E}{\rho}}$$

EACH k HAS 2 NORMAL
MODES

2 NORMAL MODES =

2 x UNIT CELLS
= 2 WAVELONGS

GENERAL

IF M ATOMS IN UNIT CELL

M MODES AT $k=0$

1 IS ACOUSTIC

$(M-1)$ ARE OPTICAL

IN 1D-2-ATOMS

2M MODES AT $k=0$

$$\begin{pmatrix} C \\ D \end{pmatrix} = \begin{pmatrix} A \\ B \end{pmatrix} e^{-ikx}$$



2 and 3 atoms

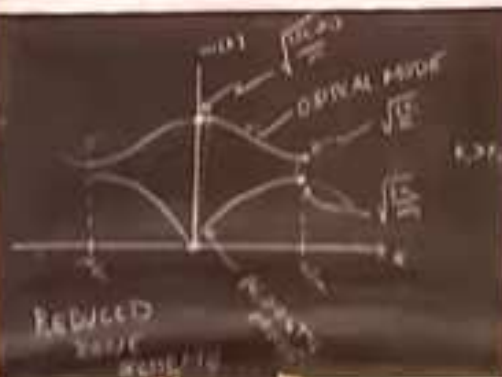
acoustic and optical



k_x, k_y MONOATOMIC CHAIN

BUT $a' = a/2$

B.R. $k = [\frac{\pi}{2a}, \frac{\pi}{2a}]$
 $= [\frac{\pi}{a}, \frac{\pi}{a}]$



GENERAL

IF M ATOMS IN UNIT CELL

M MODES AT EACH k

1 IS ACOUSTIC

$(M-1)$ ARE OPTICAL

IN 2-DIMENSIONS

$2M$ MODES AT EACH k

2 ARE ACOUSTIC

$(2M-2)$ ARE OPTICAL